Hints: Polynomials – 9. Cyclotomic polynomials and the arithmetic of roots of unity

Homework

The hints to each problem are cumulative. Read one, try again, and open the next only if you are still stuck.

Problem 1

Hint 1. Power sums: with a generator g of $\mathbb{F}_p^*$, $\sum_{x\neq 0} x^k = \sum_{j=0}^{p-2}(g^k)^j$ is a geometric sum, which vanishes unless $g^k = 1$, i.e. $(p-1) \mid k$. Don't forget the term 0^k.

Hint 2. Hermite: the number N_a of solutions of $f(x) = a$ satisfies $N_a \equiv \sum_x (1 - (f(x) - a)^{p-1}) \pmod p$. Expand $(f(x)-a)^{p-1}$ binomially; $\sum_x f(x)^t$ is minus the coefficient of x^{p-1} in $f^t \bmod (x^p - x)$. Under the conditions this gives $N_a \equiv 1$ for all a, and $\sum_a N_a = p$ forces $N_a = 1$. Conversely, for a permutation $\sum_x f(x)^t = \sum_y y^t = 0$ for $t \leqslant p-2$.

Hint 3. $x \mapsto x^k$ is injective on the cyclic group $\mathbb{F}_p^*$ iff $\gcd(k, p-1) = 1$. If $\deg f = d > 1$ and $d \mid p-1$, take $t = (p-1)/d \leqslant p-2$: f^t has degree exactly $p-1$, so it is its own remainder and the condition fails.

Problem 2

Hint 1. Compare roots: both sides are monic with simple roots, so it suffices to check that they have the same roots and degree. If $p \mid n$, x has order pn iff x^p has order n, and $\deg \Phi_n(x^p) = p\varphi(n) = \varphi(pn)$.

Hint 2. If $p \nmid n$, the roots of $\Phi_n(x^p)$ are the elements of order pn and of order n, and $p\varphi(n) = \varphi(pn) + \varphi(n)$. For odd $n > 1$, $\zeta \mapsto -\zeta$ maps primitive n-th roots onto primitive $2n$-th roots, and $(-1)^{\varphi(n)} = 1$.

Hint 3. $\Phi_{pq} = \frac{(1-x)(1-x^{pq})}{(1-x^p)(1-x^q)} \equiv (1-x)\sum_{i,j\geqslant 0} x^{ip+jq} \pmod{x^{pq}}$, and $\deg \Phi_{pq} < pq$. A number $m < pq$ has at most one representation $ip + jq$, so the sum has $0/1$ coefficients and Φ_{pq} has coefficients in $\{-1, 0, 1\}$. For Φ_{105}: modulo x^8 it equals $(1 + x + \cdots + x^7)(1 - x^3 - x^5 - x^7)$; read off the coefficient -2 of x^7.

Problem 3

Hint 1. Let $z_1, \ldots, z_n$ be the roots. The coefficients of $f_m = \prod(x - z_i^m)$ are symmetric integer polynomials in the z_i, hence integers, and are bounded by $\binom{n}{j}$ because $|z_i^m| \leqslant 1$.

Hint 2. So only finitely many polynomials f_m occur, and each root z yields the sequence $z, z^2, z^4, \ldots$ of roots of these finitely many polynomials. Hence $z^{2^a} = z^{2^b}$ for some $a < b$, and $z \neq 0$ (as $f(0) \neq 0$) gives that z is a root of unity.

Hint 3. An algebraic integer with all conjugates of modulus 1: apply this to its minimal polynomial. $\frac{3+4i}{5}$ has modulus 1, as does its conjugate, but its minimal polynomial $5x^2 - 6x + 5$ is not monic, so it is not an algebraic integer and hence not a root of unity.

Problem 4

Hint 1. A die is a polynomial $D(x) = \sum x^{\text{label}}$ with non-negative integer coefficients, $D(1) = 6$ and $D(0) = 0$. You need $D_1 D_2 = (x + \cdots + x^6)^2 = x^2(1+x)^2(1+x+x^2)^2(1-x+x^2)^2$.

Hint 2. By unique factorization each D_i is a product of these factors. $D_i(0) = 0$ gives one factor x each; $D_i(1) = 6$ with $\Phi_2(1) = 2$, $\Phi_3(1) = 3$, $\Phi_6(1) = 1$ gives exactly one $(1 + x)$ and one $(1 + x + x^2)$ each. Only the two copies of $1 - x + x^2$ can be distributed freely.

Hint 3. Split $1 + 1$: ordinary dice. Split $0 + 2$: $(1, 2, 2, 3, 3, 4)$ and $(1, 3, 4, 5, 6, 8)$ (Sicherman dice); check that the coefficients are non-negative.

Problem 5

(a)

Hint 1. Sort the positive integers by the exponent of 2: the progressions $\{2^n(2m + 1) : m \geqslant 0\}$, with first term 2^n and difference 2^{n+1}, $n \geqslant 0$.

(b)

Hint 1. Generating functions: for $|z| < 1$ a partition into progressions $a_k + d_k\mathbb{Z}_{\geqslant 0}$ gives $\sum_k \frac{z^{a_k}}{1-z^{d_k}} = \frac{z}{1-z}$.

Hint 2. Let d be the largest difference ($d \geqslant 2$) and let $z \to e^{2\pi i/d}$ radially. Only the term with $d_k = d$ blows up (for $d_k < d$, $d \nmid d_k$), while the right-hand side stays bounded. Answer: no.

Problem 6

Hint 1. P and Q are coprime (a common factor would divide P^2+1 and P), so $PQ \mid P^2+Q^2+1$. Write $P^2 + Q^2 + 1 = S\,PQ$.

Hint 2. If $\deg P = \deg Q$, then S is a constant; comparing leading coefficients gives $S = 2$ and $(P - Q)^2 = -1$, so $Q = P \pm i$.

Hint 3. If $\deg P > \deg Q$, Vieta jump: $P' = (Q^2 + 1)/P = SQ - P$ is monic, gives the same S for the pair (Q, P'), and $\deg P' < \deg Q$. Descend until the smaller polynomial is constant, hence equal to 1; this forces the other one to be 1 too and $S = 3$, contradicting $\deg S > 0$. Answer: $P = Q = 1$, or $P = R$, $Q = R \pm i$ with R monic and non-constant (in either order).

Problem 7

Hint 1. Write $m = p^a k$ with $p \nmid k$, so that $\Phi_m(x) = \Phi_{pk}(x^{p^{a-1}})$. Splitting the exponents of f by their residue mod p^{a-1} reduces to the case $a = 1$, i.e. $m = pk$ and $\deg f < p\varphi(k)$.

Hint 2. Write $f = \sum_{r<p} x^r F_r(x^p)$. The primitive pk-th roots are $\omega\eta$ with ω a primitive p-th root and η a primitive k-th root, and $(\omega\eta)^p = \eta^p$ is again a primitive k-th root. So $\sum_r \omega^{ir}\,\eta^r F_r(\eta^p) = 0$ for $i = 1, \ldots, p - 1$, which forces $\eta^r F_r(\eta^p)$ to be independent of r.

Hint 3. If fewer than p coefficients are nonzero, some F_r is 0, so all F_s vanish at every primitive k-th root, i.e. $\Phi_k \mid F_s$. A nonzero F_s then has degree $\geqslant \varphi(k)$, so $\deg f \geqslant p\varphi(k)$, a contradiction.

Problem 8

Hint 1. For a fixed N consider the polynomials $c(x) = \sum c_j x^j$ with $\sum_j c_j f(y + j/N) = 0$ for all y. They form an ideal of $\mathbb{C}[x]$, closed under gcd.

Hint 2. For a prime $p \mid N$ the hypothesis with step $1/p = \frac{N/p}{N}$ says that $\frac{x^N-1}{x^{N/p}-1}$ is in the ideal. The gcd of these over $p \mid N$ is Φ_N (Theorem 1), so Φ_N annihilates f.

Hint 3. Φ_N is monic of degree $\varphi(N)$ with constant term 1, so f at a point is determined by its values at the next (or previous) $\varphi(N)$ points of step $1/N$. Choose N (a product of many small primes) with $\varphi(N)/N < b - a$; then vanishing on (a, b) propagates by $1/N$ in both directions.

Problem 9

Hint 1. Use $k!\,(p-1-k)! \equiv (-1)^{k+1} \pmod p$ to rewrite $f(x) = \sum_k k!\,x^k$: for $x \neq 0$, $f(x) \equiv -x^{p-1} g(-1/x)$ with $g(x) = \sum_{j=0}^{p-1} x^j/j!$ in $\mathbb{F}_p[x]$. So you must bound the number of nonzero roots of g in $\mathbb{F}_p$ by $(p-1)/2$.

Hint 2. $g' = g + x^{p-1}$ (as $(p-1)! \equiv -1$). Consider $h = x^p - x + g$, of degree p.

Hint 3. At a nonzero root $r \in \mathbb{F}_p$ of g: $h(r) = 0$ and $h'(r) = -1 + g(r) + r^{p-1} = 0$, so r is a double root of h. Hence g has at most $(p-1)/2$ nonzero roots; with $f(0) = 1$ this leaves at least $(p+1)/2$ good n.

Problem 10

Hint 1. Suppose f vanishes at $\geqslant n-k+1$ of the n-th roots of unity. With $c_{n-1} = 1$, the values $f(\omega^j)$ are the discrete Fourier transform $\hat{c}$ of $c = (c_0, \ldots, c_{n-1})$, so $\hat{c}$ is supported on a set T with $|T| \leqslant k-1$.

Hint 2. The hypothesis says that the cyclic sequence $e_i = c_i\, c_{n-2-i}$ (indices mod n) is $\delta_{i,n-1}$: it vanishes for $i \leqslant n-2$, and $e_{n-1} = c_{n-1}^2 = 1$.

Hint 3. The transform of a pointwise product is a convolution, so $\hat{e}$ is supported on $T - T$, which has at most $(k-1)(k-2)+1 = k^2 - 3k + 3 < n$ elements. But the transform of δ has no zeros. Contradiction.